B

LV.0 → LV.1 · AI ENGINEER

The 6 B's that move you from Level 0 → 1 AI engineer.

Backpropagation, BERT, Bagging and more. **Plain-English** definitions, one swipe each, plus India's sovereign-AI tools and the people worth following.

BACKPROPAGATION · FOUNDATIONS

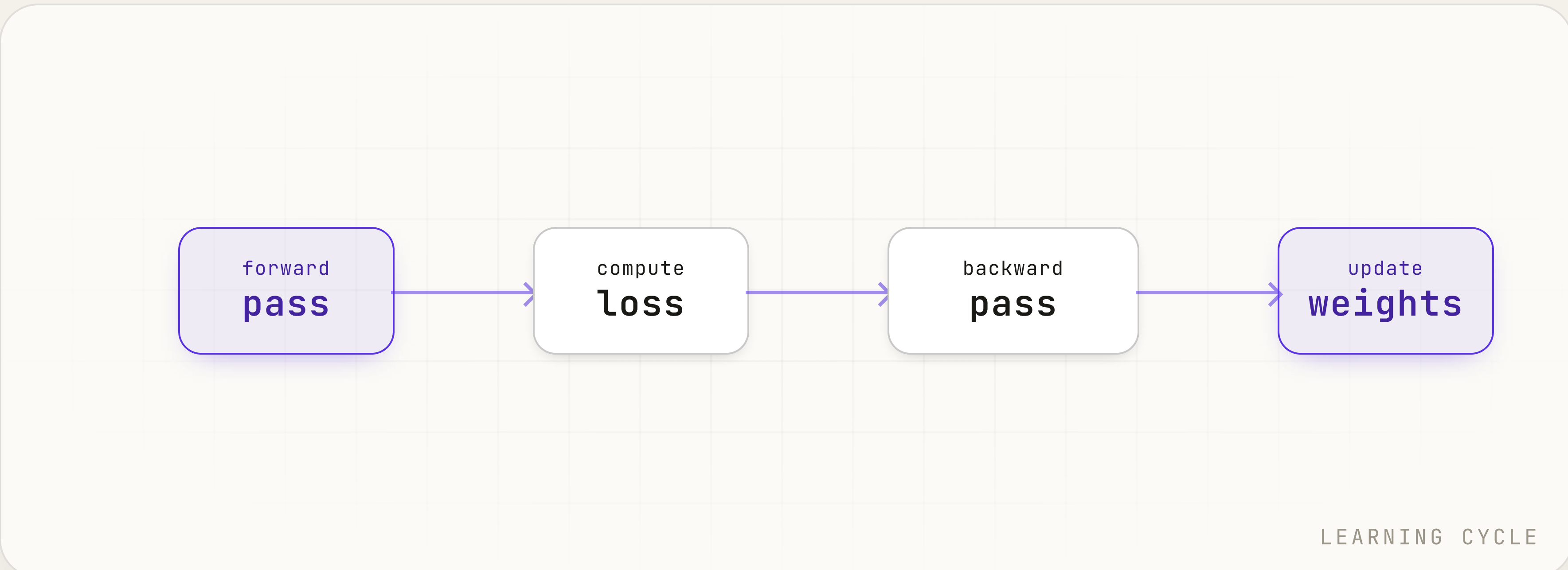
Backpropagation

error → gradient → update

Backpropagation is the fundamental learning algorithm behind neural networks. It calculates how much **each neuron contributes to prediction errors** and adjusts the model's weights using gradient descent, enabling AI systems to learn patterns, improve accuracy, and power everything from image recognition to large language models.

USED IN

- Deep learning
- Neural networks
- LLMs



BAG OF WORDS · NLP

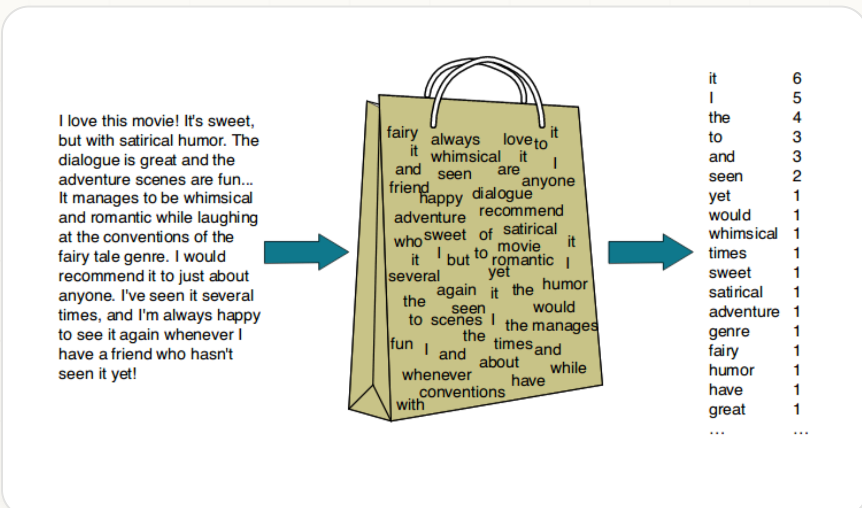
Bag of Words

word → count → vector

Bag of Words is a foundational natural language processing technique that represents text as **word-frequency vectors** while ignoring grammar and word order. Despite its simplicity, it remains a powerful baseline for text classification, sentiment analysis, spam detection, and information retrieval tasks.

USED IN

- Text classification
- Sentiment analysis
- Baseline NLP



TEXT → BAG → COUNTS

BIDIRECTIONAL ENCODER REPRESENTATIONS FROM TRANSFORMERS · TRANSFORMERS

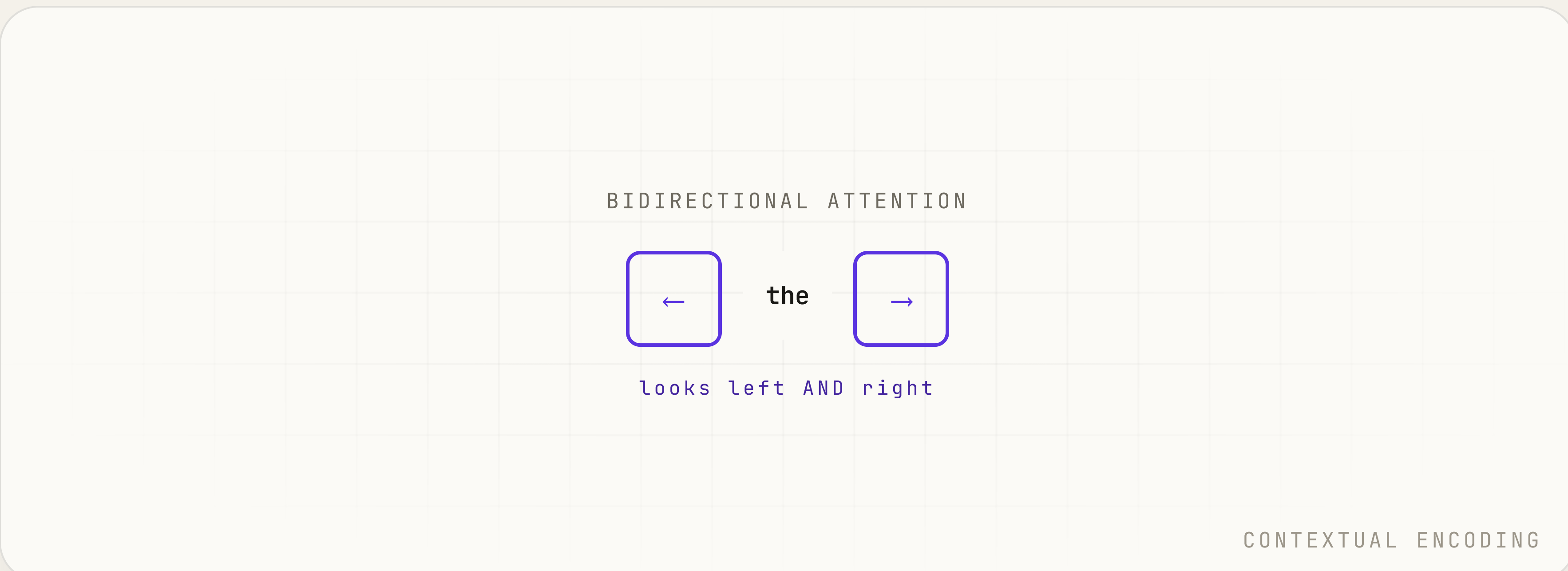
BERT

bidirectional · context · encoder

BERT is a groundbreaking language model developed by Google that understands text by analyzing context from **both directions simultaneously**. It revolutionized natural language processing, achieving state-of-the-art results in tasks such as search, question answering, sentiment analysis, and text classification across the industry.

USED IN

- Search ranking
- Q&A systems
- Text classification



BOOTSTRAP AGGREGATING · ENSEMBLE

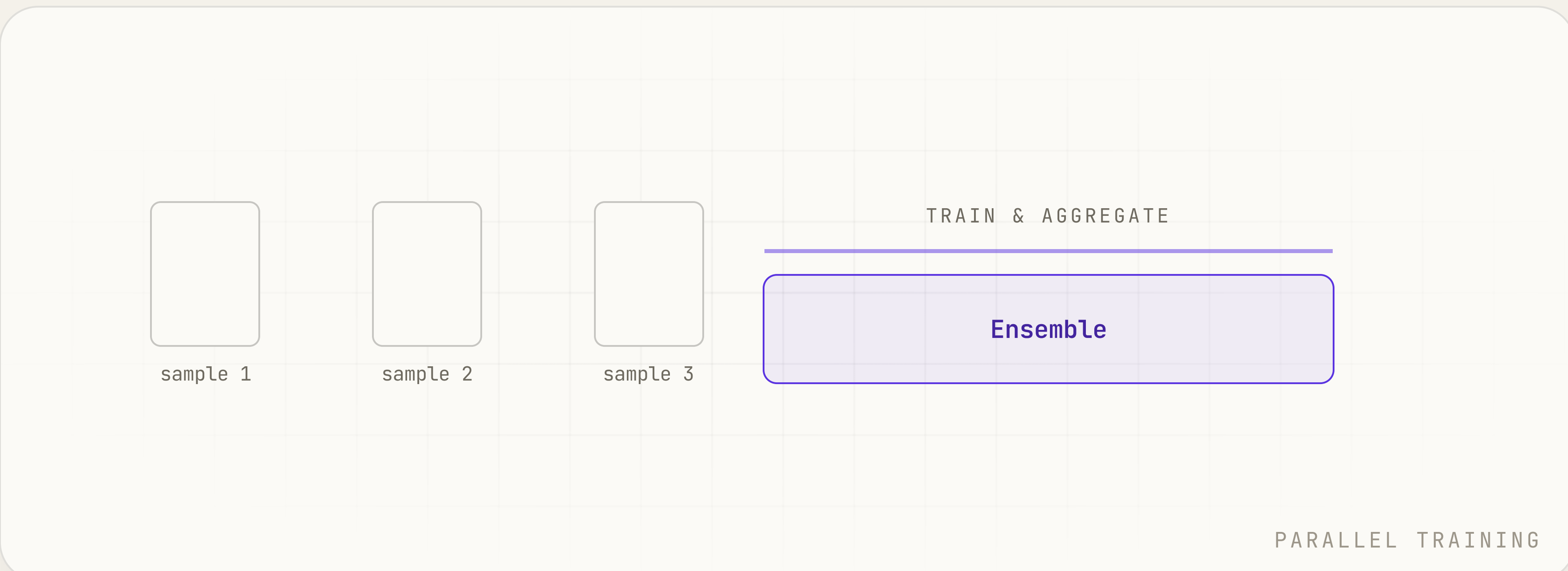
Bagging

sample · train · combine

Bagging, short for Bootstrap Aggregating, is an ensemble technique that trains multiple models on randomly sampled subsets of data and **combines their predictions**. By reducing variance and minimizing overfitting, it improves model stability, accuracy, and robustness, with Random Forest being its most well-known application.

USED IN

- Random Forest
- Variance reduction
- Robust models



SEQUENTIAL ENSEMBLE LEARNING · ENSEMBLE

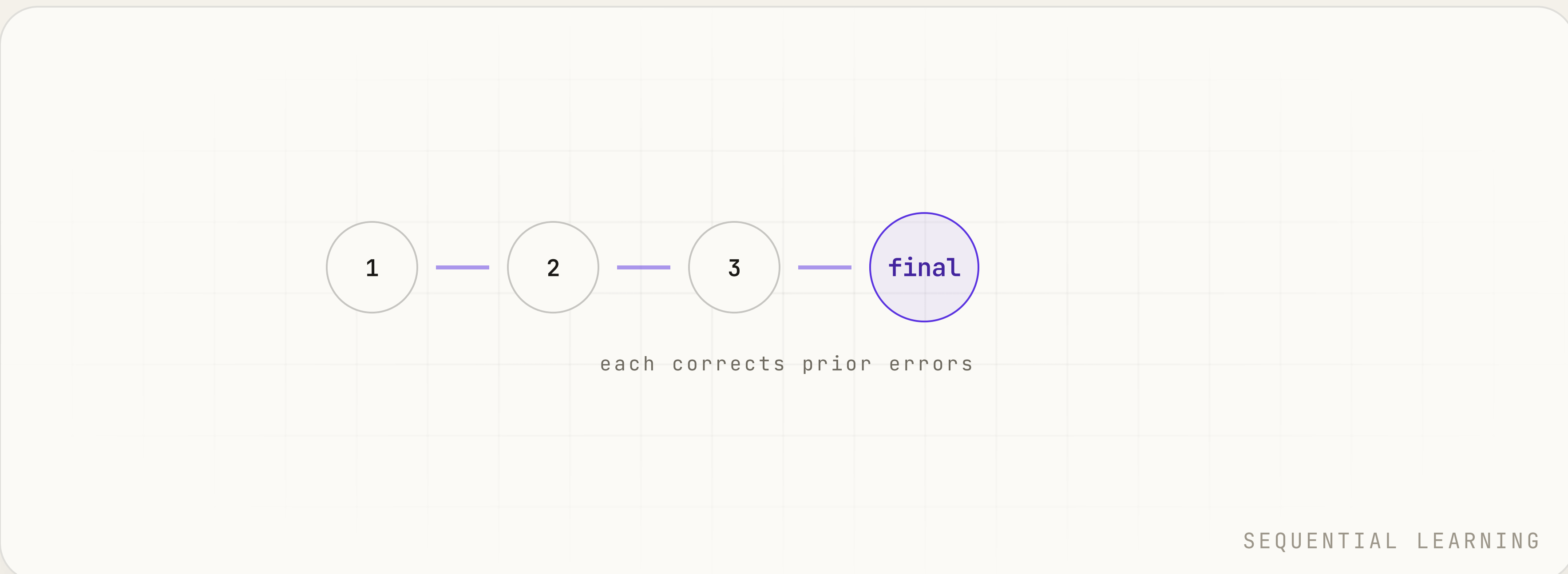
Boosting

weak → strong → sequential

Boosting is an ensemble technique that builds models sequentially, with each new model focusing on **correcting the errors of its predecessors**. By combining many weak learners into a single strong predictor, it achieves high accuracy and powers popular algorithms such as AdaBoost, Gradient Boosting, XGBoost, and LightGBM.

USED IN

- XGBoost
- Gradient Boosting
- LightGBM



DATA PREPROCESSING · DATA

Bucketing

continuous → discrete → ranges

Bucketing is a data preprocessing technique that groups continuous numerical values into **discrete ranges or intervals**. By reducing data granularity and highlighting broader patterns, it simplifies analysis, improves model interpretability, and is widely used in feature engineering, data visualization, and machine learning pipelines.

USED IN

- Feature engineering
- Data analysis
- Interpretability

AGE → BUCKETS

0–17	18–65	65+
Child	**Adult**	**Senior**

DISCRETIZATION

↓ UP NEXT · TOOLS

Tools next.

Beyond the 6 B-words sit the **India-built AI** projects breaking language barriers and building sovereign large language models.

B

LANGUAGE TRANSLATION INFRASTRUCTURE

Bhashini

National Language Translation Mission

Bhashini is India's government-backed mission that uses AI to **break language barriers** through multilingual speech, translation, and conversational technologies. By enabling seamless communication across Indian languages, it empowers citizens, developers, businesses, and public services while advancing inclusive, accessible digital innovation.

↗ bhashini.gov.in

in /company/bhashini

WHAT IT IS

India's mission for seamless multilingual access.

WHY BUILDERS USE IT

Reach 1.4B citizens in their native languages.

IN AI STACKS

Govtech, education, healthcare, and fintech across India.

PAIRS WITH

LLMs, voice interfaces, chatbots, and content systems.

SOVEREIGN FOUNDATION MODELS

BharatGen

GenAI for Bharat, by Bharat

BharatGen is India's first sovereign AI initiative, building **open, multilingual, culturally grounded LLMs** that serve every Indian. Backed by the Department of Science and Technology, it advances inclusive AI for governance, education, healthcare, agriculture, and public innovation while fostering research and talent.

in /company/bharatgen X @BharatGen_AI

WHAT IT IS

Government-backed models grounded in Indian languages.

WHY BUILDERS USE IT

Open-weight, India-first LLMs with cultural nuance.

IN AI STACKS

Governance, education, healthcare, and agriculture tools.

BACKED BY

India's Department of Science and Technology.

↓ UP NEXT · PEOPLE TO FOLLOW

Spotlight next.

The **high-signal, low-noise** people worth a follow to keep your B-stack current.

B

HIGH-SIGNAL, LOW-NOISE

BAIR

Berkeley AI Research

BAIR, or Berkeley AI Research, is **one of the world's leading academic AI labs**, bringing together UC Berkeley researchers across machine learning, computer vision, NLP, robotics, and AI safety. It is known for pioneering research, the Caffe framework, and producing influential AI founders.

↗ bair.berkeley.edu

X @berkeleybair

in /company/bair-lab

WHAT THEY POST

Research papers, open frameworks like Caffe, lab updates.

WHY FOLLOW

Their work shapes industry standards years ahead.

FIND THEM

bair.berkeley.edu, X @berkeleybair, and LinkedIn.

BEST FOR

Researchers and engineers wanting rigorous foundations.

HIGH-SIGNAL, LOW-NOISE

ByteMonk

by Himalay · system design + AI

ByteMonk, created by Himalay, simplifies system design, distributed systems, and AI through **clean visual explanations and real-world case studies**. His structured teaching helps developers move beyond interview prep toward genuinely understanding how large-scale software systems are built and scaled in production.

in /in/bytemonk   ↗ blog.bytemonk.io

WHAT THEY POST

Visual breakdowns of system design and AI.

WHY FOLLOW

Real intuition for scaling, not just buzzwords.

FIND THEM

LinkedIn @bytemonk and blog.bytemonk.io.

BEST FOR

Backend and ML engineers leveling up architecture.

HIGH-SIGNAL, LOW-NOISE

ByteByteGo

System design + AI infra, visualized

ByteByteGo is a leading software engineering education platform that simplifies **system design, distributed systems, and modern AI infrastructure** through high-quality visual explainers, newsletters, and videos. Its practical approach helps developers understand how large-scale applications are architected and scaled in the real world.

X @bytebytego in /company/bytebytego

WHAT THEY POST

Crisp architecture diagrams plus a popular newsletter.

WHY FOLLOW

A clear source on system and AI infra design.

FIND THEM

X @bytebytego and LinkedIn (ByteByteGo).

BEST FOR

Anyone prepping system-design interviews.

See you for C.

Save it, share it, and follow along. The 6 B's, India's two sovereign-AI tools, and three creators worth following.

01	Backpropagation	foundations
02	Bag of Words	nlp
03	BERT	transformers
04	Bagging	ensemble
05	Boosting	ensemble
06	Bucketing	data
+	Bhashini	tool
+	BharatGen	tool
+	BAIR	research
+	ByteMonk	creator
+	ByteByteGo	creator

FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

in/prasanna-kulkarni-032650180

C Next